VC Scout: Revised Results, Insights Summary, and Implementation Plan

Briefing note and audit-trail companion | Team 17 - Mia Murphy, Finn Kliewer, Kayvon Jafarzadeh, Nathanael Gill, Om Patel

Executive readout. The revised analysis moves VC Scout from a unicorn-only valuation exercise to a defensible benchmark-and-scouting framework. We preserve the original descriptive findings, rerun/audit the expanded data, correct funding-unit risks for unicorn rows, and separate what the data can support from what it cannot. The result is a two-layer framework: (1) a unicorn valuation benchmark and residual outperformance model, and (2) a broader tier diagnostic that improves context but is not treated as a causal startup-success model.

Traceability: revised analysis script vc_scout_final_analysis.py; source-of-truth file vc_scout_source_of_truth_final.json; final slide deliverable team17_vc_scout_revised_results_insights_implementation_final.pptx.

Traceability reference key

Tag	Audit file(s)	Used for
A1	vc_scout_audit_summary_final.csv; vc_scout_source_of_truth_final.json	dataset audit counts, funding correction counts, Form D quality checks
A2	vc_scout_audited_startup_master_final.csv; vc_scout_tier_summary_final.csv	expanded tier counts, valuation/funding coverage, corrected funding fields
A3	vc_scout_era_regime_summary_final.csv; vc_scout_unicorn_year_counts_final.csv; vc_scout_hypothesis_summary_final.csv	2021/COVID-era regime analysis and hypothesis H1
A4	vc_scout_model_results_final.csv; vc_scout_permutation_importance_final.csv	valuation benchmark model comparison and feature importance
A5	vc_scout_unicorn_residuals_final.csv; vc_scout_residuals_by_industry_final.csv; vc_scout_residuals_by_country_final.csv	residual outperformance analysis by company, industry, and geography
A6	vc_scout_tier_classifier_diagnostic_final.csv; vc_scout_formd_q2_2026_audited_final.csv	expanded tier diagnostic and Form D market-context screen

1. Refined analytics approach

Core change. The project now separates valuation benchmarking from startup scouting. The original model asked how large a unicorn valuation gets once a company is already in the unicorn population. The revised approach treats that as a benchmark problem, then uses residuals and tier comparisons as scouting signals rather than as guaranteed predictions.

- Layer 1 - Unicorn valuation benchmark: predict ln(valuation) among current/former unicorns using audited funding, industry, geography, tier, era, investor count, and years to unicorn. The output is expected valuation, not investment advice. [A4]
- Layer 2 - Scouting signal layer: compare segments by residual outperformance, sample size, sector concentration, and data-confidence flags. Companies or segments above the benchmark become candidates for deeper diligence. [A5]
- Layer 3 - Expanded-tier diagnostic: use the 75,230-row master file to compare unicorns, soonicorn proxies, accelerator controls, funded controls, exited unicorns, and delisted unicorns. This improves context but does not eliminate source bias or survivorship bias. [A2, A6]

Audit references: model comparisons are in vc_scout_model_results_final.csv; residual outputs are in vc_scout_unicorn_residuals_final.csv and segment residual tables; expanded tier diagnostics are in vc_scout_tier_classifier_diagnostic_final.csv.

2. Data shortcomings and how we addressed them

Issue	What we found	Treatment in revised analysis	Audit trail
Survivorship bias	Original Kaggle file only covered companies that reached unicorn status.	Do not claim probability of startup success from unicorn-only data; use conditional valuation and benchmark language.	A2, A4
Expanded data source bias	The master file has 75,230 rows but mixes current unicorns, old funded controls, accelerator portfolios, and proxy tiers.	Use as a diagnostic comparison universe, not a causal success model.	A1, A2
Funding-unit risk	950 unicorn rows were flagged as unit-suspect when valuation and funding were both present.	Preserve raw values and create audited funding values for analysis. After correction, 0 audited unicorn rows had funding greater than valuation.	A1, A2
Sparse valuation coverage	Only 1,829 of 75,230 expanded rows have valuation.	Limit valuation modeling to rows with defensible valuation data; avoid valuation claims for control tiers without valuations.	A1, A2
Form D noise	Q2 2026 Form D contains 5,728 rows; 238 raises exceed $100M, but 156 are likely non-startup categories.	Use Form D only as market context unless filtered to operating-company raises.	A1, A6

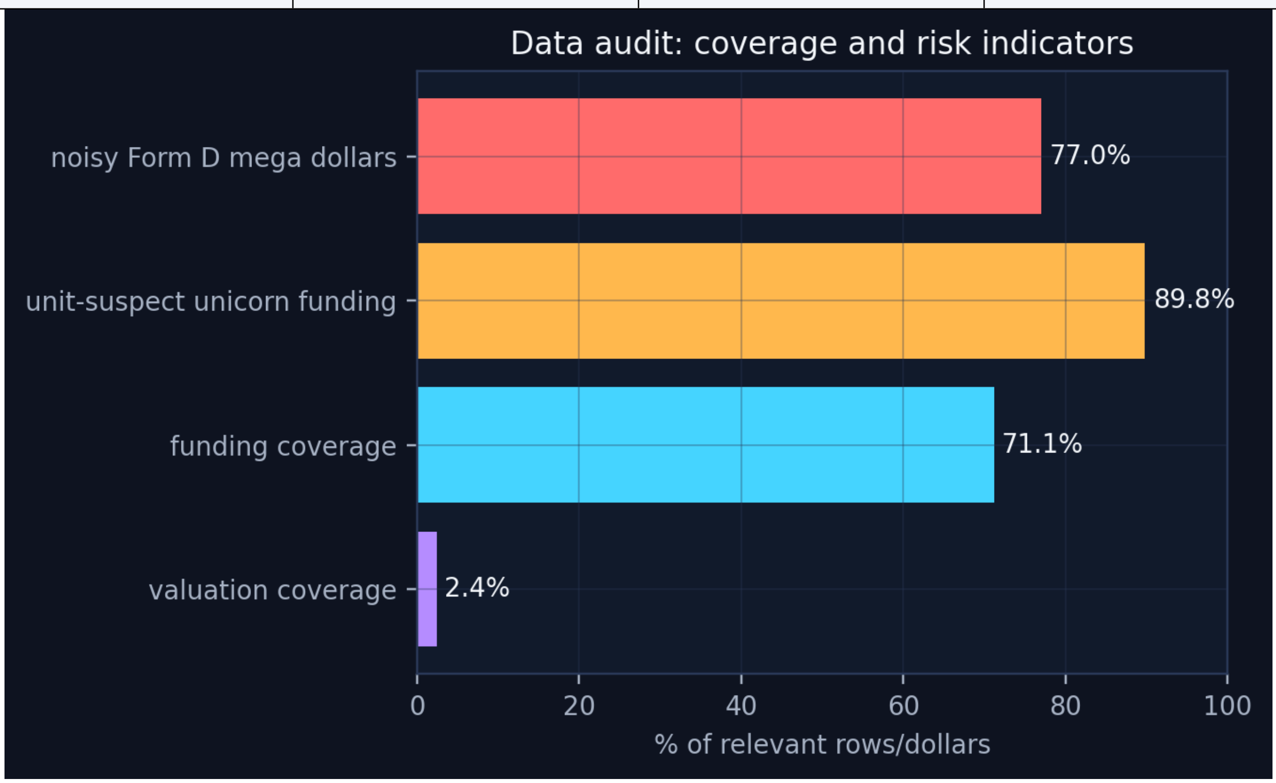

Figure 1. Audit coverage and usable data by field. Source: audit_coverage_final.png; underlying files: vc_scout_audit_summary_final.csv and vc_scout_tier_summary_final.csv. [A1, A2]

3. Detailed insights and hypotheses

Hypothesis	Status	Evidence	Implication
H1 - 2021 is a distinct market regime	Supported descriptively/statistically	2021 is the largest single dated cohort in the expanded current-unicorn list (441 rows); era differences in valuation and funding are statistically significant; sector mix differs by era.	Bring in external VC funding/interest-rate data only if the final project needs causal COVID interpretation.
H2 - Funding predicts valuation with diminishing returns	Supported	Audited benchmark champion test R2=0.30; no-funding stress test R2=0.05; log-log funding elasticity ~= 0.49.	Use benchmark residuals rather than raw valuations to spot capital-efficient outperformance.
H3 - Sector/geography add signal beyond funding	Partially supported	Industry, country, and continent residual tables show directional variation, but several medians are muted and small categories require confidence flags.	Use sample-size penalties and confidence bands before turning residuals into rankings.
H4 - Expanded controls solve survivorship bias	Disproven / only partially solved	The 75k-row master adds comparison tiers, but sources are mixed across time and valuation coverage is only 1,829 rows.	Use expanded data for diagnostic tier comparisons, not causal startup-success prediction.
H5 - Investor count is enough for investor network effects	Not proven	Investor count is a weak proxy for investor quality, timing, and co-investor networks.	Add top-fund flags and co-investor network features in final implementation.

Insight 1 - 2021 is a distinct market regime, but not automatically a COVID-causality claim

- The expanded current-unicorn list shows 441 dated unicorn rows in 2021, compared with 351 pre-2021 and 607 post-2021. Median valuation declined from $2.30B pre-2021 to $1.67B in 2021 and $1.50B post-2021; median funding also declined from $0.505B to $0.320B and then $0.231B. [A3]
- Kruskal-Wallis tests indicate significant differences by era for valuation and funding, while years-to-unicorn did not differ significantly. The sector mix also differs significantly by era. [A3]
- Interpretation: 2021 should be treated as a financing/valuation regime shock and analyzed separately. The dataset flags the discontinuity, but external VC funding, interest-rate, and digital-adoption data would be required to make a causal COVID claim. [A3]

Insight 2 - Funding matters, but with diminishing returns

- The audited valuation benchmark selected Gradient Boosting as the champion among the tested models, with test R2 = 0.30, MAE = 0.48 log units, and median absolute percentage error of about 40%. [A4]

Reconciliation with the prior model deliverable: the earlier R2 near 0.51 came from the original Kaggle-only unicorn modeling setup. The revised R2 is lower because the benchmark is rebuilt after the expanded-data audit, funding-unit corrections, and inclusion of current/former unicorn-history tiers. We prefer the lower audited estimate because it is more defensible for implementation. [A4]

- A no-funding stress test produced test R2 = 0.05, confirming that most company-level valuation signal comes from funding. [A4]
- The OLS companion model estimates funding elasticity around 0.49, which means valuation rises with funding but less than one-for-one. [A4]
- Business implication: VC Scout should not reward raw capital raised. The useful signal is whether a company, sector, or geography outperforms its expected valuation after controlling for funding. [A5]

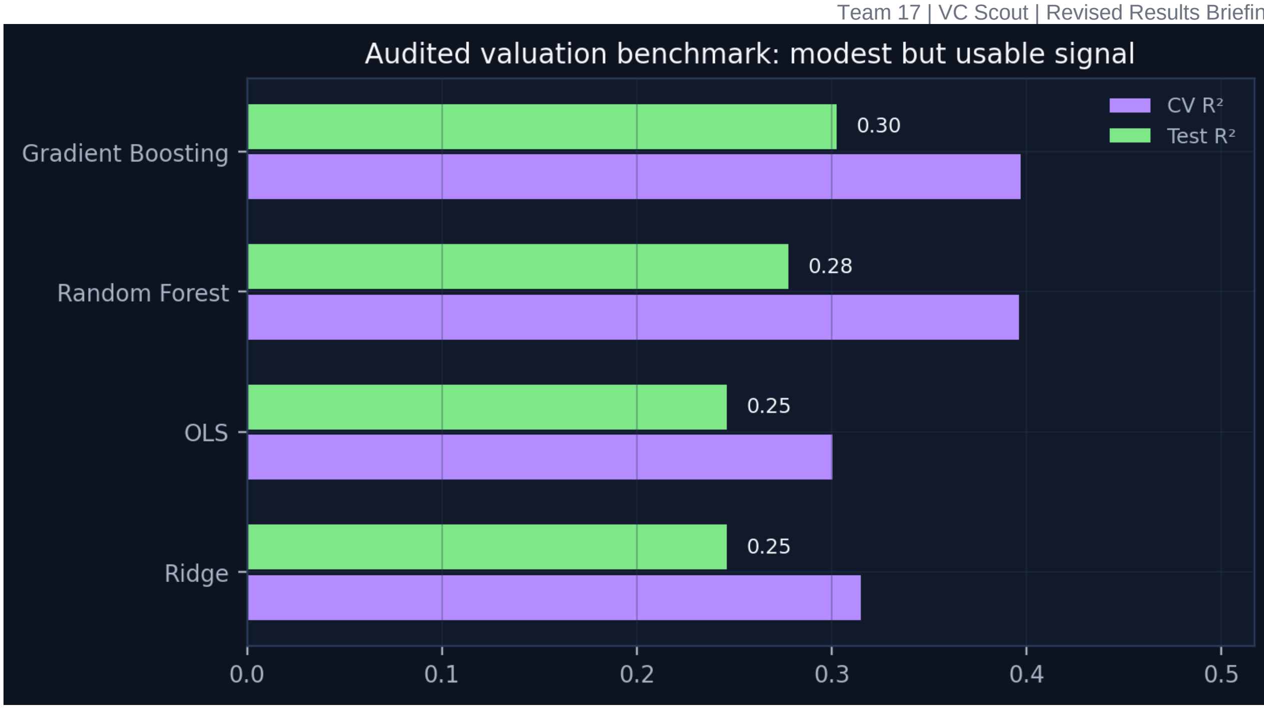

Figure 2. Audited model comparison. Source: model_results_final.png; underlying file: vc_scout_model_results_final.csv. [A4]

Insight 3 - Sector and geography add signal, but residuals are the better scouting lens

- Permutation importance shows ln(funding) is dominant, while industry and continent add smaller but positive signal. Era, tier, investor count, and years-to-unicorn are much weaker once funding is included. [A4]
- Industry residuals are directionally useful: Cybersecurity, Fintech, Enterprise Software, and Health & Bio show positive average residuals in the audited benchmark, while Transport & Logistics and E-commerce & Consumer underperform in this sample. [A5]
- Country residuals show pockets of positive benchmark outperformance, but some segments have small sample sizes. Rankings should therefore include sample-size penalties and confidence bands. [A5]

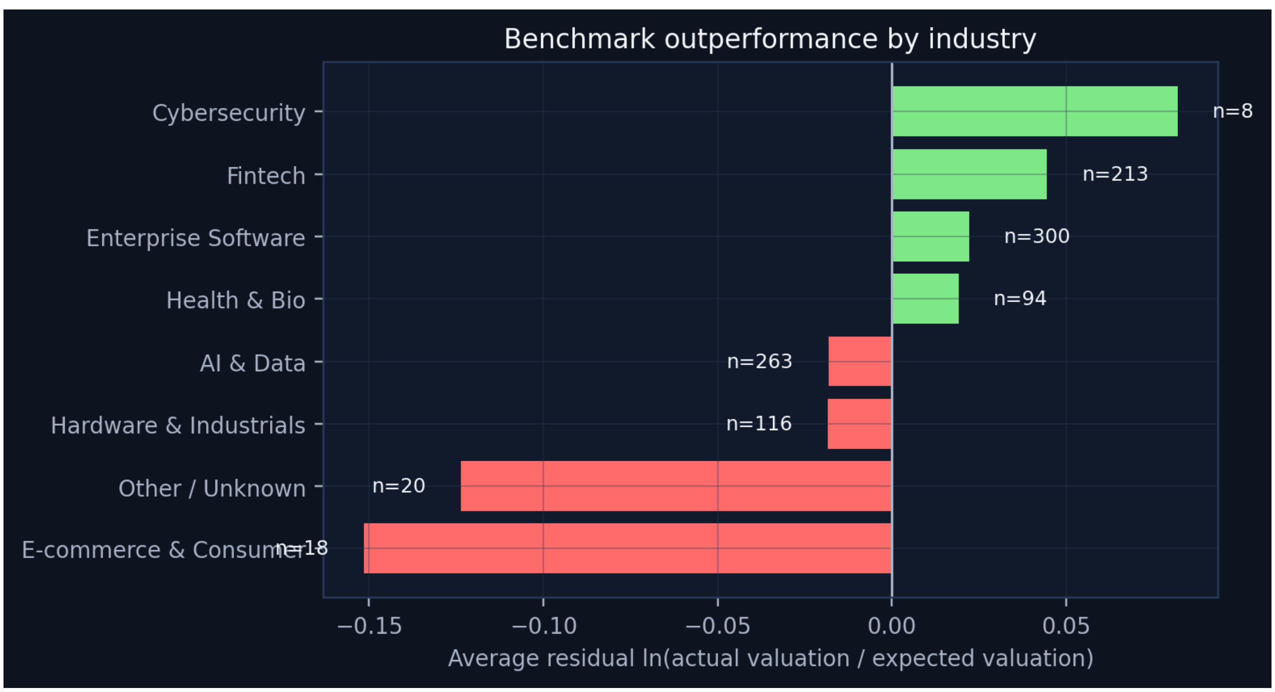

Figure 3. Industry-level valuation residuals. Source: industry_residuals_final.png; underlying file: vc_scout_residuals_by_industry_final.csv. [A5]

Insight 4 - The expanded classifier looks strong, but may be learning data-source artifacts

- The expanded-tier diagnostic model reports very high performance, including Gradient Boosting ROC AUC = 0.993 and balanced accuracy = 0.933 on a sampled tier diagnostic. [A6]
- This is useful as a diagnostic, but not a final success model. The comparison tiers are drawn from different sources and time periods, so the classifier may separate sources rather than true startup quality. [A2, A6]
- Final positioning: the expanded dataset reduces unicorn-only tunnel vision, but it only partially addresses survivorship bias. VC Scout should use it for benchmarking and triage, not causal investment decisions. [A2, A6]

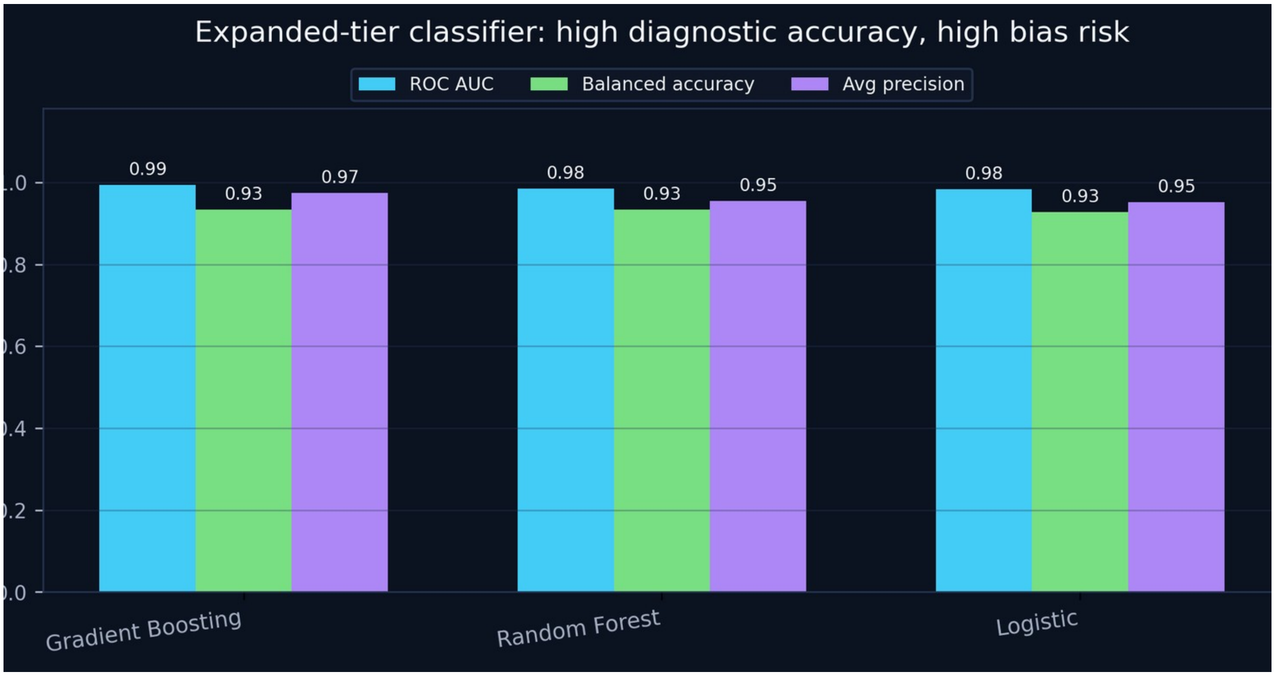

Figure 4. Expanded tier diagnostic results. Source: tier_classifier_final.png; underlying file: vc_scout_tier_classifier_diagnostic_final.csv. [A6]

4. Initial business actions and recommendations

Recommendation	Action	Reason	Traceability
Use benchmark residuals, not raw valuation	Rank sectors, countries, and investor clusters by actual minus expected log valuation.	This controls for funding and market regime instead of rewarding capital intensity.	A4, A5
Treat 2021 as its own regime	Report pre-2021, 2021, and post-2021 insights separately.	2021 is statistically distinct in valuation, funding, and sector mix.	A3
Use expanded data as context	Use tier comparisons to broaden the story, but label them diagnostic.	Mixed source timing and sparse valuation coverage prevent causal success claims.	A1, A2, A6
Add investor network features	Replace raw investor count with top-fund flags, co-investor clusters, and sector specialization.	Investor count had near-zero importance and does not capture quality or timing.	A4, A5
Publish confidence flags	Attach sample-size and confidence warnings to each sector/geography ranking.	Some high-residual countries or sectors have small n, so rankings should not imply false precision.	A5

5. Implementation plan for final capstone week

Workstream	Next deliverable	Owner logic	Source/audit dependency
Data pipeline	One reproducible cleaned dataset plus documented exclusions.	Data lead maintains raw-to-clean lineage.	vc_scout_final_analysis.py; A1, A2
Benchmark model	Locked valuation benchmark with residual exports.	Model lead owns train/test, diagnostics, and stress tests.	A4, A5
VC Scout scoring layer	Transparent score blending residuals, sector concentration, efficiency, and confidence.	Framework lead translates model outputs into decision-support language.	A3, A4, A5
Dashboard/prototype	Interactive or slide-based view by sector, geography, investor cluster, and era.	Visualization lead builds audience-ready outputs.	All chart PNGs; source-of-truth JSON
Final narrative	Recommendation deck with limitations stated clearly.	Presentation lead ensures the story is not overclaiming.	This memo plus final PPTX/PDF

Appendix - audit-trail inventory for submission

The audit trail ZIP is designed to make the work reproducible. The key files to submit with the deck and this memo are: vc_scout_audited_startup_master_final.csv, vc_scout_model_results_final.csv, vc_scout_unicorn_residuals_final.csv, vc_scout_residuals_by_industry_final.csv, vc_scout_residuals_by_country_final.csv, vc_scout_hypothesis_summary_final.csv, vc_scout_source_of_truth_final.json, and vc_scout_final_analysis.py. These files preserve the raw-to-clean logic, the exact metrics used in the deck, and the exported tables behind the visuals.

Submission package reference: vc_scout_revised_results_audit_trail_final.zip. Companion slide files: team17_vc_scout_revised_results_insights_implementation_final.pptx and .pdf.